Ghodake Raviraj Ravsaheb

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.24
Shraddha Junior Collage of Science and Commerce	Class XII (Maharashtra State Board)	2024	83.33%
New English School and Junior collage Pachegaon BK	Class X (Maharashtra State Board)	2022	97.00%

PROJECTS

CollabSpace AI - Full Stack Collaborative Workspace Platform | (Link)

Tech Stack: React.js, Node.js, Express.js, PostgreSQL, Prisma ORM, Socket.IO, Google Gemini AI, JWT

- Built a **full-stack collaborative workspace** with authentication, workspace management, rich-text editing, and document management.
- Implemented **real-time multi-user editing** and collaborator presence using **Socket.IO** with live synchronization.
- Integrated **Google Gemini** for document summarization, content generation, and AI-powered writing assistance.
- Designed **REST APIs**, **JWT authentication**, and **role-based access control** using Node.js, Express.js, Prisma, and PostgreSQL.

SmartDoc AI -- Document based answering platform using RAG | (Link)

Tech Stack: Python, FastAPI, PostgreSQL, pgvector, Ollama, Streamlit, JWT

- Built **SmartDoc AI**, a document-based question-answering system that allows users to upload PDFs and ask questions using **RAG**.
- Developed the **document processing pipeline** to extract text from PDFs, split documents into **chunks**, generate **embeddings** using Sentence Transformers, and store them in **PostgreSQL with pgvector** for similarity-based retrieval.
- Implemented a **FastAPI backend** with user authentication, PDF upload, document retrieval, and chat APIs, enabling users to securely access and query their uploaded documents.
- Integrated **Ollama with Llama 3.2 (3B)** for local AI responses and built a **Streamlit frontend** for document upload and interactive chat, including relevant document sources with page and chunk references.

RecoverAI -- AI Revenue Recovery Agent |(Link)

Tech Stack: Next.js, React, TypeScript, Tailwind CSS, Google Gemini API, Vercel

- Built an **AI-powered revenue recovery application** that analyzes failed payments, identifies root causes, and recommends recovery actions.
- Integrated **Google Gemini** for payment-failure diagnosis, recovery probability estimation, and AI-driven recommendations.
- Developed a **responsive UI** and deployed the application on **Vercel**, with secure API credential management using environment variables.

C++ Virtual File System | (Link)

- Developed a command-line virtual file system in **C++** using Object-Oriented Programming principles.
- Implemented file system operations including **mkdir, touch, ls, cd, pwd, write, cat, rm, tree,** and **find**.
- Designed the file system using a **tree data structure** with **inheritance** and **polymorphism** to represent files and directories.
- Used **recursion** for directory traversal and **file I/O** to save the virtual file system state.

SKILLS

- **Programing Languages:** C++, Python, JavaScript, SQL
- **Web Technologies:** React.js, Vite, Node.js, Tailwind CSS, bootstrap, Express.js, Vercel, Render, HTML, CSS

- **Databases:** MySQL, PostgreSQL, MongoDB
- **AI/ML:** RAG, Sentence Transformers, Gemini, Ollama
- **Tools and Technologies:** Git, GitHub, VS code
- **Computer Science Fundamentals:** Data structures and algorithm, DBMS, Object-Oriented Programming (OOP), Operating Systems

RELEVANT COURSES

- **Mathematics:** Introduction to Statistics, Introduction to Probability, Differential Equations, Complex Variables, Calculus I & II, Elementary Linear Algebra
- **Programing:** Introduction to Programing

ACHIEVEMENTS

- **Problem Solving:** Solved 300+ Data Structure and Algorithm problems across all platforms
- **Contest:** Leetcode rating- 1567, Codeforces rating-1096

EXTRACURRICULAR

National Sports Organization (NSO) - Athletics, IIT Hyderabad

- Active member of the IIT Hyderabad NSO Athletics program, fostering teamwork, consistency, and physical fitness through regular training.